MS-ML Studio: methodological core for MS preprocessing, cross-validated feature extraction, and MLP classification

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Abstract

MS-ML Studio is an open-source MATLAB application that unifies mass-spectrometry (MS) preprocessing, three complementary feature extractors with embedded K-fold cross-validation, and multilayer perceptron (MLP) classification. This note summarizes the **algorithmic pipeline**—normalization, reshaping, optional Icoshift alignment, fuzzy moving-average peak detection, correlation- and probability-based peak selection, Watch Points, Feature Resonance with genetic-algorithm (GA) optimization, and MLP training—as implemented in the MS-ML Library and GUI. An illustrative intact-cell MALDI-TOF *Aeromonas* case study [1, 2] is referenced only to anchor partitions and metrics; claims remain proof-of-concept software validation, not clinical benchmarking.

Keywords: mass spectrometry, feature extraction, genetic algorithm, MALDI-TOF, machine learning, open-source software

1. Introduction

MS workflows recur through import, preprocessing, feature extraction, and classification [3, 2, 4]. MS-ML Studio targets users who need these stages in one graphical session without scripting. The present text **focuses**

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on methods; narrative introduction, extended related work, and discussion appear in the full article under `latex/articlelsevier/`.

2. Materials and methods

2.1. Reference data (illustrative)

Spectra follow the intact-cell MALDI-TOF *Aeromonas* setting described by Surek *et al.* [1]. For each one-against-all experiment, 90% of spectra train feature extractors and the MLP; 10% are held out for blind testing (Table 1).

2.2. Architecture

The MS-ML Library implements all algorithms; the GUI orchestrates preprocessing, peak selection, feature extraction with cross-validation, and classification.

2.3. Preprocessing

2.3.1. Import and normalization

Spectra are two-column text files (m/z, intensity). Min–max normalization maps intensities to $[0, 1]$:

$$N(x, x_{\max}, x_{\min}) = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \,. \tag{1}$$

2.3.2. Reshaping

Each spectrum is binned onto W evenly spaced grid points over the global m/z range; per-bin intensity is the maximum of contributing points and m/z is the mean of their positions [5] (full discrete definition in the long manuscript).

2.3.3. Alignment

Optional alignment uses Icoshift [6] on user-selected m/z windows after reshaping.

2.3.4. Fuzzy moving average (FMA) peaks

At position p, a triangular-window local average $S(x; p, w)$ (Eq. 2) highlights local maxima as candidate peaks; w is half-width and T the triangular weight (same functional form as Watch Points below).

$$S(x; p, w) = \frac{\sum_{i=p-w}^{p+w} x_i \cdot T(2w+1, i)}{2w+1} \,. \tag{2}$$

2.4. Best peaks selection

From FMA candidates, **correlation-based** selection scores each peak's presence/absence vector against class labels (Pearson); peaks with $|\rho| > \tau$ survive. **Probability-based** selection uses class-conditional presence probabilities for discriminancy and exclusivity (see full text).

2.5. Feature extraction with cross-validation

All GA-driven extractors embed K-fold cross-validation: a configuration is retained only if the mean of training and validation correlations improves [5].

2.5.1. Watch Points

For odd window length L, triangular weights are

$$w(L,n) = \begin{cases} \dfrac{2n}{L+1} & 1 \le n \le \dfrac{L+1}{2} \\ 2 - \dfrac{2n}{L+1} & \dfrac{L+1}{2} + 1 \le n \le L \end{cases} . \tag{3}$$

(Even L uses the paired formulas in the repository manuscript.) Feature W_k for watch point k is the dot product of weights with the binary peak vector; the GA optimizes half-widths $\{L_k\}$.

2.5.2. Feature Resonance

Peaks are encoded via a sum of M sinusoids with GA-tuned parameters:

$$S(x; \mathbf{a}, \mathbf{b}, \mathbf{c}) = \sum_{i=1}^{M} a_i \sin(b_i + c_i x) \,. \tag{4}$$

Fitness is the mean absolute Pearson correlation between S and labels; variants operate on m/z grids, best-peak intensities, m/z values, or watch-point vectors.

2.6. Classification and metrics

Features from enabled extractors are concatenated and fed to an MLP (Levenberg–Marquardt). Accuracy, precision, recall, and F1 follow standard definitions [7]. ARFF export enables external classifiers (e.g. WEKA).

3. Results (summary)

Table 1: Reference partitioning (same as full manuscript).

Species	Training (n)	Test (n)	Total (n)
A. hydrophila	511	56	567
A. caviae	511	56	567
A. veronii/sobria	286	30	316
A. trota	511	56	567

Held-out test accuracies and F1-scores on this reference task are reported in the full article; the smallest test fold (*A. veronii/sobria*, $n = 30$) warrants cautious interpretation.

4. Conclusion

MS-ML Studio implements a closed pipeline from text spectra through GA-regularized feature extractors to MLP classification. This compact note documents the methodological core; extended results, discussion, limitations, and availability match `latex/articlelsevier/main.tex`.

Acknowledgements

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